SIA101 2025

SET 1 SUPERSYSTEM

STUDENT MANAGEMENT SYSTEM

SUBSYSTEM 1.1:

ENROLLMENT SYSTEM

BACKGROUND OF THE PROJECT

PROBLEM:

The primary problem that led to the development of the system is the inefficiency, disorganization, and high error rate associated with manual enrollment procedures.

SOLUTION:

The solution is an automated Enrollment System that streamlines registration, enhances data accuracy, and replaces the slow and error-prone manual process with an organized and efficient digital workflow.

GENERAL OBJECTIVE

The main objective of the proponents is to develop an enrollment system that enhances the efficiency and reliability of the enrollment process in educational institutions, ensuring a systematic and well-organized approach to student registration and record management.

SPECIFIC OBJECTIVE

1. To identify the bottlenecks encountered in the manual enrollment process in order to develop a system that streamlines registration and reduces errors.
2. To develop a system that automates students' subjects and schedules.
3. To develop a digital form that simplifies student enrollment and ensures accurate data collection.
4. To validate the freshmen's requirements for the enrollment process.
5. To generate a unique student ID number for each student, ensuring accurate identification and prevent duplication within the enrollment system, and provide a registration form.
6. To quantify the monitoring of students' status with data analytics.
7. To integrate the subsystem into the overall system by connecting it with other subsystems to create a unified super system.

SCOPE

1. Analyze manual enrollment and develop an automated system.
2. Automate subject selection, scheduling, and conflict detection.
3. Provide a digital enrollment form.
4. Validate freshmen documents.
5. Generate unique student IDs and registration forms.
6. Offer admin dashboard and enrollment statistics.
7. Integrate with learning management system and library system.

DELIMITATIONS

1. The system does not manage summer subject fees, as these are handled separately.
2. It does not offer academic advising or curriculum planning—only registration and validation.
3. It has no classroom features like attendance or grading.
4. It requires a stable internet connection since it runs online through MongoDB Atlas.
5. It is limited to only two academic programs and does not support others.

SUBSYSTEM 1.2:

LEARNING MANAGEMENT SYSTEM WITH ANTI-CHEAT

BACKGROUND OF THE STUDY

THE PROBLEM

- Online and hybrid learning increased opportunities for students to cheat due to limited supervision during digital assessments.
- Existing LMS platforms do not provide built-in monitoring to detect suspicious actions like tab switching or loss of focus during exams.
- Schools rely on multiple separate tools, making assessment monitoring difficult and inconsistent.

THE SOLUTION

LMS that centralizes class management and integrates anti-cheat tools such as kiosk mode, activity tracking, tab-switch detection, and window-focus monitoring to reduce cheating and support fair online assessments.

GENERAL OBJECTIVES

To develop a Learning Management System that centralizes class management while integrating anti-cheat mechanisms to promote fairness and integrity during online assessments.

SPECIFIC OBJECTIVES

- To develop a system that addresses the lack of monitoring features in online assessments by integrating anti-cheat mechanisms to detect and prohibit unusual behavior.
- To develop a system with anti-cheat features including kiosk full screen mode, mouse and keyboard tracking, tab switching detection, window focus monitoring, and inactivity detection for Google Form assessments.
- To develop a system that allows instructors to create classes with auto-generated codes, upload lessons and activities, manage submissions, post announcements, and upload grading sheets for Enrollment System processing.

SPECIFIC OBJECTIVES

- To design a system that allows students to join classes using class codes, access learning materials, submit outputs, view announcements, and complete Google Form assessments provided by instructors.
- To integrate a library search function that allows students to view available and reserve books through the system interface.
- To integrate the subsystem with the Library and Enrollment Systems for seamless data sharing and streamlined academic processes.

SCOPE

- The system will implement monitoring and control measures during assessments to observe students' actions and reduce opportunities for cheating.
- The system will monitor student actions in Google Form assessments using anti-cheat features, including kiosk full screen mode, activity tracking, tab switching detection, window focus monitoring, and inactivity detection to ensure assessment integrity.
- The system will enable professors to create and manage classes, upload lessons, activities, and assessments, post announcements, handle student submissions, and submit grading sheets using the prescribed template for Enrollment System processing.

SCOPE

- The system will provide students with a dashboard to join classes using class codes, access uploaded lessons and resources, submit outputs, view announcements, and take assessments embedded in Google Forms under monitored conditions.
- The system will provide a search library function where students can search, reserve, and view available books, helping them access relevant learning resources efficiently.
- The subsystem will be integrated with the Library and Enrollment Systems to synchronize student and course data, providing unified access to resources and records while maintaining existing functionalities.

DELIMITATIONS

- The system monitors student activity only during Google Form assessments and cannot prevent or log actions such as copy-paste, right-click, inspect element, or screenshots within the Google Form iframe due to Cross-Platform Limitations.
- The system cannot prevent cheating via external devices (e.g., phones, second monitors) or offline collaboration.
- Advanced proctoring features such as webcam/microphone monitoring, facial recognition, or screen recording are not included.
- The system does not calculate grades; instructors must upload grading sheets using the prescribed template for processing.

DELIMITATIONS

- The library function is limited to viewing and reserving available books; no other library management features are included.
- The system does not manage enrollment or other institutional services beyond grading sheet processing.

SUBSYSTEM 1.3:

LIBRARY SYSTEM

BACKGROUND OF THE PROJECT

The Problem:

- The library's fully manual process makes book searching slow, borrowing records can be inaccurate, and penalty tracking difficult. Students cannot reserve books in advance, and librarians struggle to manage records efficiently

The Solution:

- An integrated, web-based library system that will digitize the catalog, automate borrowing reservations, enabling online reservations and tracking of penalties

GENERAL OBJECTIVES

- To develop a Library System that replaces manual library processes with a web-based solution that manages catalogs, supports efficient borrowing and reservations, and penalty tracking

SPECIFIC OBJECTIVES

- To identify challenges in book location and search efficiency for students
- To identify access problems across different user roles (students, librarians, and admins)
- To identify inefficiencies in manual recording of borrowing and return transactions
- To design a real-time monitoring system for borrowings, overdue books, penalties, and resources
- To develop a centralized system for organized and updated library resource record
- To integrate the subsystem into the overall system by connecting it with other subsystems, creating a unified platform where the library links seamlessly with the enrollment and LMS modules.

SCOPE

- Provides a searchable catalog with real-time availability.
- Uses role-based access for students, librarians, and admins.
- The system will digitize the logging and tracking of borrowing and return transactions
- Includes dashboards for inventory, overdues, and reservations.
- Keeps a centralized and updated record of library resources.
- To integrate with Enrollment System and LMS

DELIMITATIONS

- Rare/Reference books are handled manually.
- Only officially enrolled students can borrow, no external users.
- The system track penalties, financial payments are external.
- Notifications are limited to Learning Management System (no email or SMS).
- The system excludes tuition, enrollment fees or related financial dealing.

THANK YOU

We Appreciate Your Time